Mohammad Montazeri

Department of Mechanical Engineering, Sharif University of Technology, Tehran, Iran
(+98) 911 893 1381 mos.montazeri.02@sharif.edu Mohammad-Montazeri montazeri-mohammad

Research Interests

• Robotics and Mechatronics • Biomechanics • Rehabilitation • Computer Vision and AI

Education

Sharif University of Technology **Tehran, Iran**
M.Sc. in Mechanical Engineering — GPA: 15.65/20 (3.6/4.0) *Oct. 2025 – Present*
Thesis: "Computer Vision Based Estimation of Fine Hand Motion for Pick-and-Place Tasks in Telerehabilitation"

University of Tehran **Tehran, Iran**
B.Sc. in Mechanical Engineering — GPA: 17.0/20 (3.6/4.0) *Oct. 2020 – Sep. 2025*
Thesis: "Kinematic Modeling and Redundancy Resolution of Bio-Inspired Gough-Stewart Robots via Deep Learning"

Publications

Redundancy Resolution of Rehabilitative Multi-Segment Gough-Stewart Platforms Under review
M. Montazeri, P. Namazian, and M. Tale Masouleh. Submitted to the 33rd National and 11th International Iranian Conference on Biomedical Engineering (ICBME 2026).

Kinematic Modeling of Bio-Inspired Hybrid Gough-Stewart Manipulator Dec. 2025
M. Montazeri, P. Namazian, M. Tale Masouleh, M. Nayeri, and A. Kalhor. Presented at the 13th RSI International Conference on Robotics and Mechatronics (ICRoM 2025).

Programming with Python: A Self-Study Tutorial Book Under review
Author and editor of a 170-page English-language introductory programming book.

Research Experience

Medical Robotics Laboratory **Sharif University of Technology**
Research Assistant — Supervisor: Prof. Saeed Behzadipour *2025 – Present*
- Developing a computer-vision motion-capture pipeline for fine motion detection in telerehabilitation.
- Integrating the pipeline with biomechanical models to simulate physically feasible grasps for neurological patients

Human and Robot Interaction Laboratory (TaarLab) **University of Tehran**
Research Assistant — Supervisor: Prof. Mehdi Tale Masouleh *2023 – 2025*
- Modeled the kinematics of serially connected Gough-Stewart platforms using deep neural networks.
- Developed redundancy-resolution methods for trajectory tracking and obstacle avoidance in biomimetic robots.
- Resulted in a Bachelor's thesis and two conference manuscripts

Selected Projects

Kinematic Modeling of Bio-Inspired Hyper-Redundant Gough-Stewart Robots **2025**
B.Sc. Thesis | Keras, PyTorch, Genetic Algorithm
- Trained neural networks for forward kinematics of multiple serially stacked GS platforms
- Designed a constrained redundancy-resolution algorithm for trajectory tracking and power consumption minimization.

Real-Time Motion Capture and IMU Game Controller **2024**
MediaPipe, OpenCV, Arduino
- Calibrated roll, pitch, and yaw measurements and mapped MPU-6050 tilt to a PyGame controller.
- Implemented smoothed, real-time leg-gesture recognition with MediaPipe.

Object Detection, Segmentation, and Robotic Grasp Point Generation **2024**
YOLOv5, FastSAM, Roboflow, OpenCV
- Trained detection and segmentation models and generated grasp poses from minimum-area rotated rectangles.

Robot-Type Image Classification with Convolutional Neural Networks **2024**
Python, PyTorch, ScikitLearn, CV2
- Collected and preprocessed an image-dataset and trained a CNN to classify five robot categories.

Capacitive Proximity Sensor for Liquid Level Measurement **2023**
Arduino, Proteus, Fritzing
- Designed, fabricated, optimized, and calibrated a high-sensitivity capacitive proximity sensor.

Selected Courses

- **Robotics and Mechatronics** 19.8/20
Prof. Mehdi Tale Masouleh – 2024
- **Artificial Intelligence** 17.0/20
Prof. Hakimeh Fadaei – 2023
- **Optimal Design** 18.0/20
Prof. Saeed Khodaygan – 2026
- **Medical Robotics** 17.5/20
Prof. Saeed Behzadipour – 2026
- **Spine Biomechanics** 18.3/20
Prof. Navid Arjmand – 2026
- **Automatic Control** 17.6/20
Prof. Ali Sadighi – 2023

Teaching Experience

Robotics Tutor: Sigma Educational Group — Jul. 2025 - Jul. 2026

Teaching Assistant: University of Tehran

- **Design of Machine Elements I**, Prof. A. Daneshmehr — Spring 2025
- **Robotics and Mechatronics**, Prof. M. Tale Masouleh — Fall 2024
- **Design of Machine Elements II**, Prof. A. Daneshmehr — Fall 2024
- **Numerical Calculations**, Prof. A. Fahim — Spring 2024
- **Mechanics of Materials I**, Prof. M. Mahnama — Fall 2023
- **Physics II**, Prof. S. Mohammadi — Spring and Fall 2023

Honors and Awards

- Ranked 120th among 155,000+ participants in Mathematics and 60th among 174,000+ participants in Foreign Languages in Iran's Nationwide University Entrance Exam **2020**
- Ranked 180th among 2000+ participants in Nationwide Entrance Exam for a Master's Degree in Mechanical Engineering **2025**

Technical Skills

Programming: Python, MATLAB, C++, ROS, Arduino, HTML/CSS, JavaScript, Markdown, LaTeX
Technologies/Frameworks: Git, Linux, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Django
Engineering: SolidWorks, AutoCAD, ANSYS, Abaqus, Simulink, Adams, Fritzing, Proteus

Languages and Certifications

English: IELTS 8.0 (Reading: 9.0, Listening: 9.0, Speaking: 6.5, Writing: 6.5) **Persian:** Native
Coursera: Supervised Machine Learning: Regression and Classification (2022)
TopLearn: Programming with Python (2022)

References

Dr. Saeed Behzadipour, Professor at Sharif University of Technology behzadipour@sharif.edu
Dr. Mehdi Tale Masouleh, Associate Professor at University of Tehran m.t.masouleh@ut.ac.ir
Dr. Mohammad Ali Nazari, Assistant Professor at University of Tehran manazari@ut.ac.ir